Python Learning Guide

Page 1 — Core Python Basics

Python is a beginner-friendly programming language. Variables store values, and values have types such as str (text), int (whole number), and bool (True/False).

1. Variables and Data Types

```
name = "Santhosh"
age = 25
is_learning = True
print(type(name)) # <class 'str'>
```

2. Lists and Loops

A list stores multiple items. A for loop processes each item one at a time.

```
names = ["Santhosh", "Ravi", "Priya"]
for name in names:
    print(name.upper())
```

Output: SANTHOSH, RAVI, PRIYA (each on a separate line).

3. Functions

A function is a reusable block of code. return sends a result back to the caller.

```
def add(a, b):
    return a + b

answer = add(3, 4)
print(answer) # 7
```

4. Strings

A string is text inside quotes, such as "Hello" or "Python". Methods like .upper() return a new uppercase string.

Python Learning Guide

Page 2 — Generators, join(), and Practice

5. List Comprehension vs Generator Expression

A list comprehension creates a list immediately. A generator expression produces values one at a time when requested.

```
names = ["Santhosh", "Ravi", "Priya"]

list_result = [name.upper() for name in names]
gen_result = (name.upper() for name in names)

print(type(list_result)) # list
print(type(gen_result)) # generator
print(next(gen_result)) # SANTHOSH
```

6. Using join()

```
names = ["Santhosh", "Ravi", "Priya"]
text = "\\n".join(name.upper() for name in names)
print(text)
```

The expression inside join() produces strings. join() combines them into one string, placing a newline between each value.

Output:

```
SANTHOSH
RAVI
PRIYA
```

7. Quick Practice

Try these exercises without looking at the answers:

1. Create a list of three favorite foods and print each one using a for loop.
2. Convert each food to uppercase using a generator expression.
3. Use '\\n'.join(...) to combine the uppercase food names into one string.
4. Use type() to check whether your result is a list, generator, or string.

Tip: A generator expression is not itself a string. It produces values; here each produced value is a string. The final result of join() is one string.